

Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of “Delipidation-induced porous expansion” Equation

Proof. As the delipidation process progresses, the degeneration of lipids frees up pore volume, altering the porosity, alternatively represented as void ratio. This can be modeled through the relative lipid concentrations over time as

$$\varepsilon'_{\text{layer}} = \varepsilon_{\text{layer}} + \left(\left(1 - \frac{C_{\text{lipid},\text{total}}}{C_{\text{lipid},0}} \right) \cdot \gamma \left(\frac{C_{\text{lipid},0}}{C_{\text{lipid},\text{total}}} \right) \right),$$

This equation takes the percentage of lipid that has already degenerated and multiplies it with the concentration of lipid within a specific tissue layer relative to overall tissue concentraion, using a sensitivity coefficient to assign an experimentally derivable importance. This equation can be simplified to

$$\varepsilon'_{\text{layer}} = \varepsilon_{\text{layer}} + \gamma \left(\frac{C_{\text{lipid},0} - C_{\text{lipid},\text{total}}}{C_{\text{lipid},\text{total}}} \right)$$

After assigning a condition to ensure that porosity doesn't exceed 1, the equation in terms of the final porosity becomes:

$$\varepsilon_{\text{layer}}' = \varepsilon_{\text{layer}} + \gamma \left(\frac{C_{\text{lipid},0} - C_{\text{lipid}}}{C_{\text{lipid},\text{total}}} \right), (\varepsilon_{\text{layer}}' \leq 1).$$